Maxim Van de Wynckel

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Education

Doctor of Sciences (PhD), Computer Science — *2019 - 2025*

Vrije Universiteit Brussel (Elsene, Belgium)

Thesis: Interoperable and Discoverable Indoor Positioning Systems

Master of Science (MSc), Applied Computer Science — *July, 2019*

Vrije Universiteit Brussel (Elsene, Belgium)
Graduated magna cum laude

Thesis: Indoor Navigation by Centralized Tracking

Bridging Programme, Computer Science — *2016 - 2018*

Vrije Universiteit Brussel (Elsene, Belgium)

Bachelor (Ba), Applied Computer Science — *June, 2016*

Erasmushogeschool Brussel (Anderlecht, Belgium)
Graduated magna cum laude

Thesis: eID IO - Applicatie Besturing

Employment

Vlaanderen connect. — *2025 - now*

Technical Team Lead

MVdW-Software — *2019 - now*

Software engineering consultant

V.U.B. Diving Center vzw — *2025 - now*

Board Member & Secretary

Vrije Universiteit Brussel

- **External Postdoctoral Researcher** — *2025 - now*
- **Academic and Teaching Staff** — *2019 - 2025*

ChemStream BV. — *2014 - 2018*

Software engineer (student job)

UZ Brussel — *2016*

Software developer internship (student job)

BITPOWER BV.

Software developer internship (student job) — *2015*

Software developer (student job) — *2015*

VDW-Consulting BV. — *2013 - 2014*

Software developer (student job)

Publications

Journal papers (peer reviewed)

Van de Wynckel, M., & Signer, B. (2025). OpenHPS: A Modular Framework to Facilitate the Development of FAIR Positioning Systems. Journal of Open Source Software, 10(110), 8113. https://doi.org/10.21105/joss.08113

Conference papers (peer reviewed)

Van de Wynckel, M., Valadez, I., & Signer, B. (2024). FidMark: A Fiducial Marker Ontology for Semantically Describing Visual Markers. Proceedings of The Semantic Web (ESWC 2024), 235–250. https://doi.org/10.1007/978-3-031-60635-9_14

Van de Wynckel, M., & Signer, B. (2023). SemBeacon: A Semantic Proximity Beacon Solution for Discovering and Detecting the Position of Physical Things. Proceedings of the 13th International Conference on the Internet of Things, IoT '23, 9–16. https://doi.org/10.1145/3627050.3627060

Van de Wynckel, M., & Signer, B. (2022). POSO: A Generic Positioning System Ontology. The Semantic Web – ISWC 2022, 231–247. https://doi.org/10.1007/978-3-031-19433-7_14

Van de Wynckel, M., & Signer, B. (2022). A Solid-based Architecture for Decentralised Interoperable Location Data. Proceedings of the 12th International Conference on Indoor Positioning and Indoor Navigation (IPIN'22), CEUR Workshop Proceedings, 3248. https://ceur-ws.org/Vol-3248/paper11.pdf

Van de Wynckel, M., & Signer, B. (2021). Indoor Positioning Using the OpenHPS Framework. 2021 International Conference on Indoor Positioning and Indoor Navigation (IPIN), 1–8. https://doi.org/10.1109/IPIN51156.2021.9662569

Posters (peer reviewed)

Van de Wynckel, M., & De Roock, E. (2026). OWL-SDA: A Multi-Agent System for Generating Synthetic Data to Support Ontology Evaluation. Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026. *(in press)*

Van de Wynckel, M., & Van Haute, G. (2026). ODDToolkit: An Open-Source Toolkit to Help Bridge the Developer Gap in Ontology Driven Design. Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026. *(in press)*

Van Haute, G., Goedertier, S., Fannes, P., & **Van de Wynckel, M.** (2026). Chemical Identity Lost in Regulation: A Study of Semantic Interoperability in European Chemical Substance Data. Posters, Demos, Blue Sky, and Tutorials at SEMANTiCS 2026. *(in press)*

Malaise, Y., **Van de Wynckel, M.**, & Signer, B. (2024). Towards Distributed Intelligent Tutoring Systems Based on User-owned Progress and Performance Data. Proceedings of the 2nd Solid Symposium (SoSy'24), 3947. https://ceur-ws.org/Vol-3947/short3.pdf

Van de Wynckel, M., & Signer, B. (2024). Discoverable and Interoperable Augmented Reality Environments Through Solid Pods. Proceedings of SoSy 2024 (Poster), 2nd Solid Symposium, Leuven, Belgium, May 2024, 3947. https://ceur-ws.org/Vol-3947/short2.pdf

Technical reports (not peer reviewed)

Van de Wynckel, M., & Signer, B. (2020). OpenHPS: An Open Source Hybrid Positioning System (WISE-2020-01). Vrije Universiteit Brussel. https://doi.org/10.48550/ARXIV.2101.05198

Datasets

Van de Wynckel, M., & Signer, B. (2025, May). Survey on the Privacy and Transparency of Location Data. https://doi.org/10.5281/zenodo.15564050

Van de Wynckel, M., & Signer, B. (2025). Sphero Dead Reckoning and CV Tracking Dataset. Kaggle. https://doi.org/10.34740/KAGGLE/DS/6760212

Van de Wynckel, M. (2025). Garage Positioning Dataset. Kaggle. https://doi.org/10.34740/KAGGLE/DS/6654647

Vermunicht, B., **Van de Wynckel, M.**, & Signer, B. (2023). 802.11 Managemement frames from a public location. https://doi.org/10.5281/zenodo.8003771

Hoebeke, N., **Van de Wynckel, M.**, & Signer, B. (2023). Object Tracking on a Monopoly Game Board. https://doi.org/10.5281/zenodo.7990434

Van de Wynckel, M., & Signer, B. (2021, May 9). OpenHPS: Single Floor Fingerprinting and Trajectory Dataset. https://doi.org/10.5281/zenodo.4744380

Press Articles

Les systèmes de positionnement intérieur interopérables permettent de localiser des personnes et des objets au sein des bâtiments https://nesdoo.com/les-systemes-de-positionnement-interieur-interoperables-permettent-de-localiser-des-personnes-et-des-objets-au-sein-des-batiments/

Les systèmes de positionnement intérieur interopérables peuvent déterminer l'emplacement des personnes et des objets à l'intérieur des bâtiments https://infohightech.com/les-systemes-de-positionnement-interieur-interoperables-peuvent-determiner-lemplacement-des-personnes-et-des-objets-a-linterieur-des-batiments/

VUB researcher teaches systems that determine the location of people or objects within buildings to work together https://press.vub.ac.be/vub-researcher-teaches-systems-that-determine-the-location-of-people-or-objects-within-buildings-to-work-together

VUB-onderzoeker leert systemen die de locatie van mensen of objecten in gebouwen bepalen, samenwerken https://press.vub.ac.be/vub-onderzoeker-leert-systemen-die-de-

locatie-van-mensen-of-objecten-in-gebouwen-bepalen-samen-werken

Interoperable indoor positioning systems can determine locations of people or objects inside buildings https://techxplore.com/news/2025-07-interoperable-indoor-positioning-people.html

Researcher Maxim Van De Wynckel Defends Ph.D. on Indoor Positioning Systems https://www.lifetechnology.com/blogs/life-technology-technology-news/researcher-maxim-van-de-wynckel-defends-ph-d-on-indoor-positioning-systems

Jong Talent: Maxim Van de Wynckel (30) zoekt oplossingen om gebruikers meer controle te geven over hun locatiegegevens https://www.knack.be/nieuws/technologie/jong-talent-maxim-van-de-wynckel-30-zoekt-oplossingen-om-gebruikers-meer-controle-te-geven-over-hun-locatiegegevens/

Presentations

Public PhD Defence

2025-06-30

Venue: Vrije Universiteit Brussel, LIC0.04, Brussels, Belgium
Public PhD defence titled 'Interoperable and Discoverable Indoor Positioning Systems'

Presentation at FOSDEM 2025

2025-02-01

Venue: FOSDEM 2025, Brussels, Belgium
Presentation titled 'Discovering Indoor Environments and Positioning Systems'

Conference presentation at ESWC 2024

2024-05-28

Venue: The Semantic Web. ESWC 2024, Hersonissos, Crete, Greece
Paper presentation titled 'FidMark: A Fiducial Marker Ontology for Semantically Describing Visual Markers'

Poster presentation at SoSy 2024

2024-05-02

Venue: The 2nd Solid Symposium, Leuven, Belgium
Poster presentation titled 'Discoverable and Interoperable Augmented Reality Environments Through Solid Pods'

Conference presentation at IoT 2023

2023-11-08

Venue: The 13th International Conference on the Internet of Things, Nagoya, Japan
Paper presentation titled 'SemBeacon: A Semantic Proximity Beacon Solution for Discovering and Detecting the Position of Physical Things'

Poster presentation at BeJS conf 2023

2023-05-12

Venue: The Belgian JavaScript Conference (BeJS conf), Brussels, Belgium
Poster presentation of the OpenHPS framework

Conference presentation at ISWC 2022

2022-10-26

Venue: The 21st International Semantic Web Conference, Hangzhou, China
Paper presentation titled 'POSO: A Generic Positioning System Ontology'

Conference presentation at IPIN 2022

2022-02-06

Venue: The 12th International Conference on Indoor Positioning and Indoor Navigation, Beijing, China
Paper presentation titled 'A Solid-based Architecture for Decentralised Interoperable Location Data'

Presentation at FOSDEM 2022

2022-02-06

Venue: FOSDEM 2022, Brussels, Belgium
Presentation titled 'Rapid Prototyping of a Positioning System Using the OpenHPS Framework'

Conference presentation at IPIN 2021

2021-12-01

Venue: The 11th International Conference on Indoor Positioning and Indoor Navigation, Lloret de Mar, Spain
Paper presentation titled 'Indoor Positioning Using the OpenHPS Framework'

Teaching Experience

Courses

Next Generation User Interfaces (4018166FNR)
2019 - 2025

Level: master **Titular:** Prof. Dr. Beat Signer

Web Technologies (1019888BNR)
2019 - 2025

Level: bachelor **Titular:** Prof. Dr. Beat Signer

Databanken (1007156ANR)
2019 - 2025

Level: bachelor **Titular:** Prof. Dr. Bas Ketsman

Web Technologies (4019862ANR)
2019 - 2023

Level: master **Titular:** Prof. Dr. Beat Signer

Databases (4019814ENR)
2019 - 2022

Level: master **Titular:** Prof. Dr. Bas Ketsman

Advanced Topics in Big Data (4023256FNR)
2021 - 2022

Level: master **Titulars:** Prof. Dr. Beat Signer, Prof. Dr. Bas Ketsman

Advanced Topics in Information Systems (4016792ENR)
2019 - 2021

Level: master **Titular:** Prof. Dr. Beat Signer

Information Visualisation (4019538FNR)
2019 - 2021

Level: master **Titular:** Prof. Dr. Beat Signer

Open Information Systems (4020612FNW)
2020 - 2021

Level: master **Titulars:** Prof. Dr. Bart Bogaerts, Prof. Dr. Bas Ketsman

Open Information Systems (4020612FNW)
2019 - 2020

Level: master **Titular:** Prof. Dr. Christophe Debruyne

Supervised Theses

LiDAR Indoor Map Matching for Autonomous Robots
2023 - 2024

Student: Jordan Shittu

Identifying Devices Based on MAC Layer Properties
2022 - 2023

Student: Benjamin Vermunicht

Heuristic Real-Time Trajectory Prediction Using Spatio-temporal Databases
2022 - 2023

Student: Ahmed Barhdadi

Location Tracking on a Physical Game Board
2022 - 2023

Student: Nathan Hoebeke

End-User Authoring for Location-based Implicit Human-Computer Interaction — *2022 - 2023*

Student: Stoffel Oostvogels

Wi-Fi Fingerprinting: Signal Visualisation and Crowd-sourcing — *2020 - 2021*

Student: Ahmed Barhdadi

Indoor Positioning by Visible Light Communication — *2020 - 2021*

Student: Ronald Michiels

Anonymous Contact Tracing Fusion — *2020 - 2021*

Student: Yassine Bouagaz

Comparison of SLAM Solutions with Exclusive Usage of LIDAR Observation Data — *2020 - 2021*

Student: Ben Bergs

Design Guidelines for IoT Systems in Smart Companies — *2020 - 2021*

Student: Jeroen Grégoire

Thesis Jury Member

Contextual Note-taking in Scholarly Workflows — *2025 - 2026*

Student: Elliott Octave

WebPresenter: A Customisable Presentation Control — *2025 - 2026*

Student: Rob Leemans

ExcerptManager: A Local-first Plugin for Excerpt-Level Scholary Information Management Based on the Resource-Selector-Link Metamodel — *2025 - 2026*

Student: Zakaria Rabahi

AI-Based Email Classification and REcommendation Using the 4D Inbox Zero Framework — *2024 - 2025*

Student: Seifallah Youssef

Dynamic Data Physilication Based on Magnetic Modularity and Power Line Communication — *2024 - 2025*

Student: Sunidhi Singh

Paper Explorer: A Multi-Pane Solution for Literature and Citation Exploration — *2024 - 2025*

Student: Mable Rajan

A Knowledge Graph-Driven Platform for Sharing Knowledge in Data Physicalisation — *2024 - 2025*

Student: Lysa Roxana Kwizera

Rethinking File Management: Designing a Human-Centric File System Interface — *2024 - 2025*

Student: Talha Jafar

A Monocular SLAM Solution for the Formula Student Driverless Competition — *2024 - 2025*

Student: Jan Boonants

Academic Services

Advisory member of the PhD of Jan Boonants — *2025 - present*

Advisory member of the PhD thesis of Jan Boonants.

Reviewer of the Semantic Web - Interoperability, Usability, Applicability (Journal) — *2025*

Reviewer of the Interactive, Mobile, Wearable and Ubiquitous Technologies (Journal) — *2022*

Reviewer of the Proceedings of the 15th International Conference on Tangible, Embedded and Embodied Interaction — *2020*

OAP representative of the Faculty Research Comittee — *2020 - 2025*

Other academic staff (OAP) representative of the faculty of science and engineering.

OAP representative of the Department Council — *2019 - 2025*

Other academic staff (OAP) representative of the department of computer science.

OAP representative of the Bachelor Program Committee — *2019 - 2025*

Other academic staff (OAP) representative of the Bachelor program of computer science.